

## C1 past final worksheet

1. (Spring 2022, #8) A car is driving on the road and suddenly notices an obstacle 500 feet away and immediately starts braking. The car's velocity in ft/s is measured every four seconds and is given in the table below

|          |    |    |    |    |    |    |
|----------|----|----|----|----|----|----|
| <b>t</b> | 0  | 4  | 8  | 12 | 16 | 20 |
| <b>v</b> | 60 | 50 | 20 | 12 | 4  | 0  |

- (a) Roughly at what time does the car come to a stop? Justify your answer.
- (b) Give an upper and a lower estimate for the distance traveled between  $t = 0$  and  $t = 20$  seconds. Justify your answer.
- (c) Based on your results, which of the following statements can you make? Please explain your answer briefly below.
- The car hits the obstacle.
  - It does not hit the obstacle.
  - We can not tell from just the given data.

2. (Fall 2020, #1) Write the following limit as a definite integral. You may use the properties of logarithms to help with your algebra. Do not evaluate the integral.

$$\lim_{n \rightarrow \infty} \frac{1}{n} \ln \left( \left(1 + \frac{1}{n}\right)^3 \left(1 + \frac{2}{n}\right)^3 \cdots \left(1 + \frac{n}{n}\right)^3 \right)$$

3. (Spring 2020, #6) Consider the integral  $\int_0^1 \sqrt{1+x^2} dx$ .

(a) Express the integral as a left Riemann sum with 4 sub-intervals of equal width. You may leave your answer as an unsimplified sum.

(b) Is the above Riemann sum in (a) greater than or less than the value of the integral? Briefly explain.

4. (Fall 2019, #3) Consider the integrals  $I = \int_0^2 (1 + x^2) dx$  and  $J = \int_0^2 e^{x^2} dx$ .

Note: the value of the integral  $J$  cannot be evaluated explicitly.

- (a) Write  $I$  as the limit of a right Riemann sum.

- (b) Compute the limit of the Riemann sum. **Hint:**  $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$ .

- (c) Given that  $e^{x^2} \geq 1 + x^2$  for all values of  $x$ , determine which value is larger:  $J$  or 4.59?

5. (Fall 2006, #8)

(a) Write the following limit as an integral.

$$\lim_{n \rightarrow \infty} \frac{2}{n} \sum_{i=1}^n \sqrt{4 - \left(\frac{2i}{n}\right)^2}$$

(b) Compute the limit by interpreting the integral as an area.